Vinitha Gadiraju

vinitha.gadiraju@wellesley.edu
linkedin.com/in/vinithagadiraju
vinithagadiraju.com

Faculty Position

Wellesley College July 2023 - Present
Assistant Professor in Computer Science, Tenure-Track

Education

University of Colorado, Boulder (CU Boulder) August 2018 - May 2023
Ph.D. Computer Science
M.S. Computer Science

University of Oregon September 2015 – June 2018
B.S. Computer Science, Minor in Psychology

Awards

2025 **Trustworthy AI Research Fellow**, Computing Research Association and Microsoft ($17,500)
2024 **Research Scholar Program Award**, Google Research ($60,000)
2022 **Outstanding Service Award**, CU Boulder, Computer Science Department ($600)
2021 **Sickle Cell Disease Challenge 1st Place Winner**, National Heart, Lung, and Blood Institute
Designed a trivia game prototype on Figma to help patients learn about Sickle Cell ($25,000)
2021 **Publication Recognition Award**, CU Boulder, Computer Science Department ($200)
2021 **Collegiate Award Finalist**, National Center for Women in Technology
2020 **Department Summer Research Fellowship for Outstanding PhD TA**, CU Boulder, Computer Science Department ($6,000)
2020 **Chateaubriand Fellow**, Embassy of France in the United States (€7,298)
2020 **Graduate Research Fellow (NSF GRFP)**, National Science Foundation ($114,000)
2020 **Collegiate Award Finalist**, National Center for Women in Technology
2020 **Award Winner for Aspirations in Computing**, National Center for Women in Technology
2019 **Graduate Student Research Competition Winner**, ACM ASSETS 2019 ($500)
1st Place Winner at the ACM Graduate Student Research Competition [P.2]

Publications * indicates authorship by an undergraduate researcher whom I advised.

Peer-Reviewed Conference Papers

[C.7] Alison Ingalls*, Kalei Kieu*, Calvin Liang, Taslima Akter, Dhruv Jain, **Vinitha Gadiraju.** 2027. Towards Responsible AI Relationship Support: Consequences of Mediation Blur and

Asymmetry. Currently under review - conditional acceptance at *CHI Conference on Human Factors in Computing Systems (CHI '27)*.

[C.6] Wenhan Xue*, Kelly Ramos Pereira*, Taniyah Hicks*, Maitraye Das, **Vinitha Gadiraju**. 2026. Navigating New Disability as a Sociotechnical Process. In Proceedings of the 2026 *SIGACCESS Conference on Computers and Accessibility* (ASSETS '26, *30% acceptance rate*; Honorable mention, *top 5.6% of submissions*). ACM. DOI: https://doi.org/10.1145/3797867.3829064

[C.5] **Vinitha Gadiraju**, Lucia Jayne, Shaun K. Kane. 2024. "It's an independent living skill, but covered with fun!": Prompting At-Home Skill Development for Children with Vision Impairment. In Proceedings of the 2024 *SIGACCESS Conference on Computers and Accessibility* (ASSETS '24, *28% acceptance rate*). ACM. DOI: https://doi.org/10.1145/3663548.3675626

[C.4] **Vinitha Gadiraju**, Shaun Kane, Sunipa Dev, Alex Taylor, Ding Wang, Robin Brewer, and Emily Denton. 2022. "I wouldn't say offensive but...": Disability-Centered Perspectives on Large Language Models. In Proceedings of the *2023 ACM Conference on Fairness, Accountability, and Transparency* (FAccT '23). ACM. DOI: https://doi.org/10.1145/3593013.3593989

[C.3] **Vinitha Gadiraju**, Jérémie Garcia, Shaun K. Kane, Anke M. Brock. 2021. "It is fascinating to make these beasts fly": Understanding Visually Impaired People's Motivations and Needs for Drone Piloting. In Proceedings of the 2021 *SIGACCESS Conference on Computers and Accessibility* (ASSETS '21, *29% acceptance rate*). ACM. DOI: https://doi.org/10.1145/3441852.3471219

[C.2] **Vinitha Gadiraju**, Olwyn Doyle*, and Shaun K. Kane. 2021. Exploring Technology Design for Students with Vision Impairment in the Classroom and Remotely. In Proceedings of the 2021 *CHI Conference on Human Factors in Computing Systems (CHI '21, 26.3% acceptance rate)*. ACM. DOI: https://doi.org/10.1145/3411764.3445755

[C.1] **Vinitha Gadiraju**, Annika Muehlbradt, and Shaun K. Kane. 2020. BrailleBlocks: Computational Braille Toys for Collaborative Learning. In Proceedings of the 2020 *CHI Conference on Human Factors in Computing Systems (CHI '20, 24.3% acceptance rate)*. ACM. DOI: https://doi.org/10.1145/3313831.3376295

Journal Articles

[J.1] Amy Ouyang, Manasa Gadiraju, Veda Gadiraju, Landon Power, **Vinitha Gadiraju**, Gloria Liu, Kristin P. Guilliams, Michael M. Binkley, Sherif M. Badawy, and Melanie E. Fields. "GRAPES: Trivia game increases sickle cell disease knowledge in patients and providers and mitigates healthcare biases." *Pediatric Blood & Cancer* 69, no. 7 (2022): e29717.

Workshop Papers

[W.1] Tiffany Horter*, Serena Booth, **Vinitha Gadiraju**. 2024. Helping People Predict Agent Behaviors by Operationalizing the Variation Theory of Learning. *NeurIPS 2024 Workshop on Behavioral Machine Learning*. https://openreview.net/forum?id=1mO4BuJowX

Posters, Extended Abstracts, and Presentations

[P.5] Wenhan Xue*, Manasa Kudumu*, Kayla Mullen*, Sneha Sriram*, Alicia Boyd, **Vinitha Gadiraju**. 2025. "Characterizing Uses and Prompting Strategies of LLM-Based Chatbots Among Neurodivergent Individuals. 2025 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '25). ACM.

[P.4] Kayla Mullen*, Wenhan Xue*, Manasa Kudumu*. Advised by **Vinitha Gadiraju**. 2024. "I'm treating it kind of like a diary": Characterizing How Users with Disabilities Use AI Chatbots. Student Research Competition (2nd Place Winner) at the 2024 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '24). ACM. DOI: https://doi.org/10.1145/3663548.3688549

[P.3] **Vinitha Gadiraju**. 2021. Tactile Play and Literacy Learning for Visually Impaired Children. Doctoral Consortium at the 2021 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '21). ACM.

[P.2] **Vinitha Gadiraju**. 2019. BrailleBlocks: Braille Toys for Cross-Ability Collaboration. In Proceedings of the 2019 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '19), (pp. 688-690). ACM. DOI: https://doi.org/10.1145/3308561.3356104

[P.1] **Vinitha Gadiraju** and Shaun K. Kane. 2019. Computational Tools to Enable Learning Braille. Poster at *CRA-W Grad Cohort for Women*.

Invited Talks

2026 Universidade de Lisboa
Mediating Care and Connection: Designing Technology for Relational Work

2026 Union College (Computer Science Seminar Speaker Series)
From Representation to Reality: How Technology, AI, and Disability Shape Each Other

2025 University of Toronto (COMPASS Accessibility Panel)

2025 University of Maryland (BBL Speaker Series)
Title: Navigating Bias and Leveraging AI: Exploring the Dual Reality for Users with Disabilities

2024 Northeastern University (Guest Lecture)
Title: Participatory Design Methods to Investigate Accessible Tools and Disability Representation

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Academic Research

Principal Investigator, Gadiraju Lab
Wellesley College, July 2023 - Current

- Designing perinatal information tools to support shared decision making between pregnant patients, their clinicians, and families. (In collaboration with Dr. Maitraye Das at Northeastern University).
  (Submitted for 2026 NSF CAREER Award, currently under review)
    - Built community partnerships and conducted preliminary grounding interviews with *Partner to Decide*, a nonprofit supporting perinatal decision-making, and *Bump and Beyond*, a pregnancy education initiative at Boston Medical Center
        - Interviews contextualized patient information-seeking within perinatal care for socioeconomically, linguistically, and culturally diverse populations. These collaborative sessions also resulted in design requirements for context-adaptive information systems
    - Conducted two workshops with the Patient Advisory Board members from *Bump and Beyond* and *Partner to Decide* to better characterize the decision types, timelines, and facets of decision-making that patients require information and support with during pregnancy.
    - Currently developing methodology and prototypes for patient workshop sessions to understand how contextual information presentation adaptations in technology (interaction modality, depth, organization, patient preparation, and accessibility) could better support decision making for diverse pregnant people.

- Developing safer AI relationship support, mediation, and counseling. (In collaboration with Dr. Calvin Liang at Northwestern University, Dr. Dhruv Jain at University of Michigan and Dr. Taslima Akter at University of Texas San Antonio).
    - Interviewed 15 people in long-term, cohabiting relationships to examine how AI chatbots mediate relational work, identifying mediation blur across emotional processing, communication, conflict resolution, and companionship. Examined consequences of asymmetric AI use for shared relationship dynamics and developed design considerations for responsible relational mediation. [C.7]
    - Currently co-designing design probes to investigate user and professional perspectives on design recommendations for relational AI support.

- Building and translating mental state visualizations for mental health support tools. (In collaboration with Dr. Yaniv Yacoby at Wellesley College).
    - Developed a “clean language” methodology to elicit unbiased visualizations of mental distress alongside Symbolic Modeling experts James Lawley and Penny Tompkins
    - Surveying and running focus groups with Wellesley College undergraduate students to build a taxonomy of common visualizations to represent emotional and mental state changes using a design tool (FigJam).
    - Developing a prototype application for adolescents in therapy to build mental model visualization using the identified metaphor elements.

- Examined how recently disabled adults navigate early disability experiences across fragmented U.S. healthcare systems, digital information tools, and online support communities. (In collaboration with Dr. Maitraye Das at Northeastern University).
    - Conducted semi-structured interviews with 14 recently diagnosed disabled adults to identify the credibility and coordination labor involved in securing care and navigating evolving disability identities, informing HCI research and design considerations for technologies that support patient navigation, reduce patient labor, and integrate clinical information with lived experience [C.6].

- Characterized Large Language Model use among neurodivergent people within their daily routines [P.4] [P.5].
    - Investigated how neurodivergent adults use LLM-based chatbots in real-world contexts through semi-structured interviews and diary studies.
    - Identified and categorized diverse chatbot use cases, examined users' prompting and response-refinement strategies, and explored chatbot strengths, limitations, and opportunities to better represent disability experiences in LLM development.

Graduate Researcher, Superhuman Computing Lab
CU Boulder, Fall 2018 - May 2023

- Investigated collaborative Independent Living Skill Development for blind or visually impaired (BVI) children and their parents at home.
    - Conducted an interview and diary study to characterize how families develop independence at home, what additional skills they value, and how technology can support development in these multifaceted areas.

- Led an ethnographic study at the Colorado School for the Deaf and the Blind to understand educator values for BVI children, current technology use for instruction, and how game-based technology can support instructors in the classroom and remotely [C.2].
    - Observed learning, teaching, and collaboration strategies between peers and educators related to Braille education, literacy, and Expanded Core Curriculum.
    - Conducted informal interviews with students, educators, and school administration.
    - Co-designed with educators to create games and activities to teach literacy and social skills for remote education and in-person instruction.

- Designed and created BrailleBlocks, a learning tool using tangible blocks and games to help visually impaired children and sighted collaborators learn, teach, and practice Braille [C.1].
    - Built tangible blocks representing Braille cells and coded an interface with corresponding educational games.
    - Programmed with OpenCV to color detect pegs in the blocks for real-time Braille letter and word detection.
    - Conducted user studies with sighted parents and visually impaired children to evaluate the system and elicit feedback.

Chateaubriand Research Fellow
École nationale de l'aviation civile, Spring 2021 – Summer 2021

- Led two projects to explore accessible drone piloting for BVI adults.
    - Surveyed and interviewed visually impaired participants to understand their motivation to fly and suggestions for how piloting can be made accessible [C.3].
    - Built a wearable accessible drone piloting system with audio and haptic feedback to indicate direction and depth during flight.
    - Conducted participatory design based user studies with visually impaired adults for early evaluation of the prototype.

Industry Experience

Research Intern, Google Research - People + AI Research
Mountain View, California, Summer 2022

- Led a project on characterizing Large Language Model harm and bias towards people with disabilities by facilitating interaction between participants and an associated chatbot [C.4]
    - Designed AI prompts to moderate conversations about disability between participants and the language model.
    - Moderated focus groups with 56 participants identifying across diverse disabilities.
    - Conducted iterative thematic analysis to generate a codebook for open coding.

Teaching

Spring 2026	Introduction to Programming (CS111), Wellesley College Human-Computer Interaction (CS220), Wellesley College
Fall 2025	Introduction to Programming (CS111), Wellesley College Designing for Accessibility (CS325), Wellesley College
Spring 2025	Introduction to Programming (CS111), Wellesley College Human-Computer Interaction (CS220), Wellesley College
Fall 2024	Introduction to Programming (CS111), Wellesley College Designing for Accessibility (CS325), Wellesley College
Spring 2024	Introduction to Programming (CS111), Wellesley College
Fall 2023	Introduction to Programming (CS111), Wellesley College Human-Computer Interaction (CS220), Wellesley College
Fall 2021	Course Manager Fundamentals of Human-Computer Interaction (CSCI 3200), CU Boulder

Fall 2018 - Spring 2020 Teaching Assistant
Fundamentals of Human-Computer Interaction (CSCI 3200), CU Boulder

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Mentorship

While at Wellesley College

Lab Members

*indicates students that participated in the Wellesley Science Center Summer Research Program

Fall 2026 Alison Ingalls, Anisa Rodriguez, Achieng Mellisa, Ivy Wang, Manasa Kudumu, Imaan Jivraj, Ziyou Li, Yuchen Xiao

Summer 2026* Alison Ingalls, Kalei Kieu, Taniyah Hicks, Anisa Rodriguez, Achieng Mellisa, Ziyou Li, Yuchen Xiao

Spring 2026 Wenhan Xue, Manasa Kudumu, Emmie Regan, Taniyah Hicks, Ivy Wang, Achieng Mellisa, Imaan Jivraj, Yuchen Xiao, Shale Tenerelli

Fall 2025 Sneha Sriram, Wenhan Xue, Manasa Kudumu, Emmie Regan, Rachel Claire Henry, Taniyah Hicks, Ivy Wang, Achieng Mellisa

Summer 2025* Taniyah Hicks, Kelly Ramos Pereira, Wenhan Xue, Ivy Wang, Achieng Mellisa

Spring 2025 Sneha Sriram, Wenhan Xue, Manasa Kudumu, Kamora Adejo, Emmie Regan

Fall 2024 Sneha Sriram, Wenhan Xue, Manasa Kudumu

Summer 2024* Wenhan Xue, Manasa Kudumu, Kayla Mullen

Spring 2024 Sneha Sriram, Wenhan Xue

Honors Thesis Advisees

2023-2024 Tiffany Horter
CS Thesis: Getting In Your Robot's Head: Building Mental Models of Robots Using Human-Concept Learning [W.1]

Honors Thesis Committees

2025-2026 Karen Xiao
CS Thesis: Quantifying Diversity of Educational Evaluation Using Numerical Scores

Aubrey Cantrell
CS Thesis: Muse or Maker: The Role of Generative AI in Artistic Collaboration

2024-2025 Janavi Padala
CS Thesis: It's Time To Chime In! A Video Game on Political Misinformation and Individual Impact

Anna Zhou
CS Thesis: Investigating How Technologies and Platforms Shape the VTuber Experience

Maya Mau

CS Thesis: Building and Assessing a Web-based, Accessible, Three-Dimensional Modeling Tool to allow High School Students to Explore Molecular Structure, Intermolecular Interactions, and Molecular Dynamics Simulations

2023-2024 Zhihan Xu
CS Thesis: The Mysteries of Large Language Models: Tracing the Evolution of Transparency in OpenAI's GPT Models

Other Student Mentoring

2024 Jami Johnson, Web Development Advising

Non-Wellesley Research Advising

2026 Gabriella Ramirez, Doctoral Student at Northwestern University
Project: Safer AI Relationship Support

2024-2025 Akankshita Dash, Graduate Student at Stanford University
Project: Wikipedia Image Descriptions for BLV Users

2024 Sarah Adams, Undergraduate Student at Harvard University
Project: Clinician Perspectives on ML in suicide risk prediction

2024 Sydney Levy, Graduate Student at Harvard University
Project: Clinician Perspectives on ML in suicide risk prediction

2023-2025 Alicia Boyd, Postdoctoral Researcher at New York University
Project: Disability Representation in Large Language Models

<u>While at CU Boulder</u>

2022-2023 Lucia Jayne - M.S. in Computer Science (CU)
2022 Alexis Grémont - M.S. in Aeronautics (École nationale de l'aviation civile)
2019 - 2021 Olwyn Doyle - B.A. in Political Science and Government and Computer Science (CU)

Service

2026 Program Committee, ACM FAccT '27, Porto, Portugal
2026 Area Chair, ACM CHI '27, Pittsburgh, USA
2026 Program Committee, ACM FAccT '26, Montréal, Canada
2026 Hybrid Chair, ACM ASSETS '26, Porto, Portugal
2025 Reviewer, ACM CHI '26, Barcelona, Spain
2025 Hybrid Chair, ACM ASSETS '25, Denver, USA
2025 Program Committee, ACM ASSETS '25, Denver, USA
2024 Area Chair, Accessibility and Aging Track, ACM CHI '25, Yokohama, Japan
2024 Juror, Experience Reports, ACM ASSETS '24, St. John's, Newfoundland and Labrador

2024 Reviewer, ACM CSCW '24, San José, Costa Rica
2024 Reviewer, ACM IDC '24, Delft, Netherlands
2024 Program Committee, ACM FAccT '24, Rio De Janeiro, Brazil
2023 Area Chair, Accessibility and Aging Track, ACM CHI '24, Honolulu, USA
2023 Reviewer, ACM ASSETS '23, New York, USA
2022 Late-Breaking Work Program Committee, ACM CHI '23, Hamburg, Germany
2022 Reviewer, ACM CHI '23, Hamburg, Germany
2022 Internship Guide Panelist, CU Boulder
2021 Teaching Assistant Hiring Interviewer, CU Boulder
2021 Late-Breaking Work Program Committee, ACM CHI '22, New Orleans, USA/Virtual
2021 Student Volunteer, ACM ASSETS, Virtual
2020 Prospective Faculty Student Host, CU Boulder
2020 Graduation Application Feedback Program Mentor, CU Boulder
2020 Student Volunteer, ACM CHI (Canceled due to COVID-19), Honolulu, USA
2019 Reviewer, ACM/IEEE HRI '20, Cambridge, UK

Organizations and Departmental Service

Environmental Sustainability Committee
Wellesley College, Fall 2024 – Spring 2026

- Member of Advisory Board to Wellesley College on fostering a more sustainable and waste minimizing campus. Participated on a sub-committee focused on reducing dining hall waste and a sub-committee promoting e-recycling in the Science Center.

WOC Faculty Mentoring Group
Wellesley College, Fall 2023 – Present

- Professional support/development group for tenure-track women of color at Wellesley.